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United States Patent	12407598
Kind Code	B2
Date of Patent	September 02, 2025
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Connectivity between virtual datacenters

Abstract

Some embodiments provide a method that receives (i) definition of a group of virtual datacenters and (ii) addition of at least two virtual datacenters to the group. Each virtual datacenter is defined in a public cloud and includes a set of network management components and a set of network endpoints connected by a logical network that is managed by the network management components of the virtual datacenter. Based on the definition of the group, the method configures a gateway router to which each of the virtual datacenters of the group connect. The gateway router is for routing traffic between the virtual datacenters of the group. The method also configures, at each respective virtual datacenter, a respective router to route data traffic between the respective virtual datacenter and the other virtual datacenters to route traffic for the other virtual datacenters to the gateway router.

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Appl. No.: 18/119208

Filed: March 08, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230239238 A1	Jul. 27, 2023

Related U.S. Application Data

continuation parent-doc US 17212662 20210325 US 11606290 child-doc US 18119208

Publication Classification

Int. Cl.: H04L45/586 (20220101); H04L45/02 (20220101); H04L45/302 (20220101);
H04L45/42 (20220101); H04L67/567 (20220101)

U.S. Cl.:

CPC H04L45/586 (20130101); H04L45/02 (20130101); H04L45/3065 (20130101);
H04L45/42 (20130101); H04L67/567 (20220501);

Field of Classification Search

CPC: H04L (45/586); H04L (67/567); H04L (45/02); H04L (45/3065); H04L (45/42)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8830835	12/2013	Casado et al.	N/A	N/A
8964767	12/2014	Koponen et al.	N/A	N/A
9137052	12/2014	Koponen et al.	N/A	N/A
9209998	12/2014	Casado et al.	N/A	N/A
9288081	12/2015	Casado et al.	N/A	N/A
9444651	12/2015	Koponen et al.	N/A	N/A
9755960	12/2016	Moisand et al.	N/A	N/A
9876672	12/2017	Casado et al.	N/A	N/A
9935880	12/2017	Hammam et al.	N/A	N/A
10091028	12/2017	Koponen et al.	N/A	N/A
10193708	12/2018	Koponen et al.	N/A	N/A
10735263	12/2019	McAlary et al.	N/A	N/A
10754696	12/2019	Chinnam et al.	N/A	N/A
10931481	12/2020	Casado et al.	N/A	N/A
11005710	12/2020	Garg et al.	N/A	N/A
11005963	12/2020	Maskalik et al.	N/A	N/A
11171878	12/2020	Devireddy et al.	N/A	N/A
11212238	12/2020	Cidon et al.	N/A	N/A
11240203	12/2021	Eyada	N/A	H04L 63/20
11362992	12/2021	Devireddy et al.	N/A	N/A
11582147	12/2022	Raman et al.	N/A	N/A
11588819	12/2022	Wei	N/A	H04L 67/1097
11606290	12/2022	Patel et al.	N/A	N/A
11729094	12/2022	Arumugam et al.	N/A	N/A
11729095	12/2022	Sadasivan et al.	N/A	N/A
2007/0058604	12/2006	Lee et al.	N/A	N/A
2008/0159150	12/2007	Ansari	N/A	N/A
2009/0003235	12/2008	Jiang	N/A	N/A
2009/0296713	12/2008	Kompella	N/A	N/A
2009/0307713	12/2008	Anderson et al.	N/A	N/A

2011/0126197	12/2010	Larsen et al.	N/A	N/A
2011/0131338	12/2010	Hu	N/A	N/A
2012/0054624	12/2011	Owens, Jr. et al.	N/A	N/A
2012/0110651	12/2011	Biljon et al.	N/A	N/A
2013/0044641	12/2012	Koponen et al.	N/A	N/A
2013/0044751	12/2012	Casado et al.	N/A	N/A
2013/0044752	12/2012	Koponen et al.	N/A	N/A
2013/0044761	12/2012	Koponen et al.	N/A	N/A
2013/0044762	12/2012	Casado et al.	N/A	N/A
2013/0044763	12/2012	Koponen et al.	N/A	N/A
2013/0044764	12/2012	Casado et al.	N/A	N/A
2013/0142203	12/2012	Koponen et al.	N/A	N/A
2013/0185413	12/2012	Beaty et al.	N/A	N/A
2013/0283364	12/2012	Chang et al.	N/A	N/A
2014/0282525	12/2013	Sapuram et al.	N/A	N/A
2014/0334495	12/2013	Stubberfield	370/401	H04L 49/354
2014/0376367	12/2013	Jain et al.	N/A	N/A
2015/0113146	12/2014	Fu	709/226	H04L 47/805
2015/0193246	12/2014	Luft	N/A	N/A
2016/0105392	12/2015	Thakkar et al.	N/A	N/A
2016/0127202	12/2015	Dalvi et al.	N/A	N/A
2016/0170809	12/2015	Schmidt et al.	N/A	N/A
2016/0182336	12/2015	Doctor et al.	N/A	N/A
2016/0234161	12/2015	Banerjee et al.	N/A	N/A
2017/0033924	12/2016	Jain et al.	N/A	N/A
2017/0063673	12/2016	Maskalik et al.	N/A	N/A
2017/0195517	12/2016	Seetharaman et al.	N/A	N/A
2017/0353351	12/2016	Cheng	N/A	H04L 12/4625
2018/0270308	12/2017	Shea et al.	N/A	N/A
2018/0287902	12/2017	Chitalia et al.	N/A	N/A
2018/0295036	12/2017	Krishnamurthy et al.	N/A	N/A
2018/0332001	12/2017	Redondo Ferrero	N/A	H04L 61/5014
2019/0068500	12/2018	Hira	N/A	N/A
2019/0104051	12/2018	Cidon et al.	N/A	N/A
2019/0104413	12/2018	Cidon et al.	N/A	N/A
2019/0149360	12/2018	Casado et al.	N/A	N/A
2019/0149463	12/2018	Bajaj et al.	N/A	N/A
2019/0327112	12/2018	Nandoori et al.	N/A	N/A
2019/0342179	12/2018	Barnard et al.	N/A	N/A
2020/0235990	12/2019	Janakiraman	N/A	H04L 41/0893
2021/0067375	12/2020	Cidon et al.	N/A	N/A
2021/0067439	12/2020	Kommula et al.	N/A	N/A
2021/0067468	12/2020	Cidon et al.	N/A	N/A
2021/0075727	12/2020	Chen et al.	N/A	N/A
2021/0112034	12/2020	Sundararajan et al.	N/A	N/A
2021/0126860	12/2020	Ramaswamy et al.	N/A	N/A
2021/0136140	12/2020	Tidemann et al.	N/A	N/A

2021/0184898	12/2020	Koponen et al.	N/A	N/A
2021/0314388	12/2020	Zhou et al.	N/A	N/A
2021/0336886	12/2020	Vijayasankar et al.	N/A	N/A
2021/0359948	12/2020	Durrani et al.	N/A	N/A
2022/0094666	12/2021	Devireddy et al.	N/A	N/A
2022/0311707	12/2021	Patel et al.	N/A	N/A
2022/0311714	12/2021	Devireddy et al.	N/A	N/A
2022/0377009	12/2021	Raman et al.	N/A	N/A
2022/0377020	12/2021	Sadasivan et al.	N/A	N/A
2022/0377021	12/2021	Sadasivan et al.	N/A	N/A
2023/0006920	12/2022	Arumugam et al.	N/A	N/A
2023/0006941	12/2022	Natarajan et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101977156	12/2010	CN	N/A
111478850	12/2019	CN	N/A
2013026050	12/2012	WO	N/A
2022060464	12/2021	WO	N/A
2022250735	12/2021	WO	N/A

OTHER PUBLICATIONS

Non-Published Commonly Owned U.S. Appl. No. 18/235,869, filed Aug. 20, 2023, 50 pages, VMware, Inc. cited by applicant

Non-Published Commonly Owned U.S. Appl. No. 18/235,874, filed Aug. 20, 2023, 69 pages, VMware, Inc. cited by applicant

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Background/Summary

CLAIM OF BENEFIT TO PRIOR APPLICATIONS (1) This application is a continuation application of U.S. patent application Ser. No. 17/212,662, filed Mar. 25, 2021, now published as U.S. Patent Publication 2022/0311707. U.S. patent application Ser. No. 17/212,662, now published as U.S. Patent Publication 2022/0311707, is hereby incorporated by reference.

BACKGROUND

(1) More and more enterprises have moved or are in the process of moving large portions of their computing workloads into various public clouds (e.g., Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure, etc.). If an enterprise uses one of the native offerings of these clouds, and only uses one public cloud for their workloads, then many of these public clouds offer connectivity between the workloads in the different datacenters of the same public cloud provider. For instance, AWS virtual private clouds (VPCs) can be connected via a transit gateway. (2) However, enterprises might want to use workloads in multiple public clouds as well as their on-premises/branch datacenters that are not as easily all connected without traffic traveling through the public Internet. In addition, some enterprises might want to retain management of their workloads

in the cloud, rather than running native cloud provider managed workloads. Better connectivity between these workloads would therefore be beneficial.

BRIEF SUMMARY

(3) Some embodiments provide a method for providing connectivity between virtual datacenters defined and operating within public clouds. The virtual datacenters, in some embodiments, are datacenters that include network and/or compute management components operating in the public cloud as well as network endpoints (workloads) connected by a logical network within the virtual datacenter. The network endpoints as well as the logical network are managed by the management components that operate within the virtual datacenter.

(4) To provide connectivity between these virtual datacenters, a cloud management platform enables a user (e.g., a network administrator of the enterprise to which the virtual datacenters belong) to define a connectivity group of virtual datacenters and add virtual datacenters to the group. In some embodiments, all of the virtual datacenters in a group are defined and operate within physical datacenters of the same public cloud provider, while in other embodiments the virtual datacenters may be defined in datacenters of multiple different public cloud providers. In addition, some embodiments impose a restriction that all of the virtual datacenters in a group must be located in datacenters within a single predefined geographic region (e.g., Eastern US, Western US, Europe, East Asia, etc.). Other embodiments allow for a group to include virtual datacenters from multiple such geographic regions.

(5) In some embodiments, the virtual datacenters are defined within virtual private clouds (VPCs) of the public cloud. A VPC, in some embodiments, is a set of workloads that are allocated to the tenant of the public cloud (e.g., an enterprise) and that are isolated from workloads of other tenants. In some embodiments, for a virtual datacenter, the tenant VPC is allocated a set of physical host computers of the public cloud that only host workload data compute nodes (e.g., virtual machines (VMs), containers, etc.) that are part of the tenant virtual datacenter (i.e., the physical host computers are not shared with other tenants of the public cloud). Within the VPC, a tenant logical network is defined, to which both the management components and the endpoint workloads connect. In some embodiments, the typical virtual datacenter logical network includes a first logical router (which can be referred to as a tier 0 (T0) logical router) for handling traffic entering and exiting the virtual datacenter as well as multiple additional logical routers that connect to this first logical router. These additional logical routers (which can be referred to as tier 1 (T1) logical routers) isolate different sets of logical switches (also referred to as network segments) from each other. Some embodiments include a first T1 logical router to which one or more management logical switches connect as well as a second T1 logical router to which one or more workload logical switches connect. The compute and/or network management components of the virtual datacenters are attached to the management logical switches. In some embodiments, the workload data compute nodes (DCNs) are connected to the workload logical switches as well as to one or more management logical switches (i.e., via different interfaces). These DCNs exchange data traffic with each other as well as DCNs at other virtual datacenters via the workload logical switches, and exchange control traffic with the management components in their virtual datacenter via the management logical switches.

(6) When a group is defined, some embodiments automatically define and configure a centralized gateway router for connecting the virtual datacenters, as well as routers at the edges of each of the virtual datacenters that connect to the gateway router. In some embodiments, this centralized gateway router is implemented as a horizontally scalable regional logical router. That is, the centralized gateway router can be implemented as a single device or scaled to multiple devices that act as a single router in some embodiments.

(7) The gateway router is configured to route data traffic between the network endpoints at each of the virtual datacenters, as well as to route management traffic between the management components and/or network endpoints at the virtual datacenters. The routers at each virtual

datacenter are configured to route traffic sent from their respective virtual datacenter and directed to network addresses at any of the other virtual datacenters to the gateway router, which acts as a hub for routing traffic between the datacenters. Correspondingly, when the gateway router sends data traffic for a particular virtual datacenter to the router at the edge of that virtual datacenter, that router routes the traffic into the logical network of its virtual datacenter. In some embodiments, the router at a particular virtual datacenter that connects to the centralized gateway router connects to the logical network at the particular virtual datacenter via the TO logical router of that logical network.

(8) In order to configure these routers that handle the inter-datacenter traffic, the cloud management platform generates routing tables and provides these routing tables to the routers in some embodiments. The routing table for the centralized gateway router includes routes that instruct the gateway router to send traffic to each of the routers at the edge of the virtual datacenters. For each virtual datacenter, the routing table of the centralized gateway router includes one or more routes specifying that traffic directed to network addresses associated with the logical network at that virtual datacenter is to be forwarded to the router at the edge of that virtual datacenter. In addition, at each virtual datacenter, the router at the edge of that virtual datacenter is configured with (i) one or more routes specifying that traffic directed to network addresses associated with the logical network at that virtual datacenter is to be forwarded to the logical network (e.g., to the T0 router) at that virtual datacenter and (ii) one or more routes specifying that traffic directed to network addresses associated with the logical networks at the other virtual datacenters is to be forwarded to the centralized gateway router.

(9) In some embodiments, the cloud management platform also allows the user to connect the virtual datacenters in a group to (i) native VPCs in a public cloud and/or (ii) on-premises datacenters. These native VPCs, in some embodiments, are not virtual datacenters in that they do not include management components and, in many cases, are not implemented on fully isolated hardware (e.g., the host computers that host DCNs of the VPC may also host DCNs for other VPCs of other public cloud tenants). The native VPCs may be managed through the cloud management platform or through a cloud provider platform. In some embodiments, the native VPCs may host services (e.g., load balancers, deep packet inspection, DHCP service, authentication services, endpoint services, etc.) used to process data traffic sent to or from the virtual datacenters.

(10) An administrator might want the virtual datacenters to be able to connect to physical on-premises datacenters if only a portion of the enterprise's workloads are moved into the public cloud. In some embodiments, an additional gateway router (e.g., a specialized on-premises connection gateway) is defined in the public cloud in order to connect the on-premises datacenter to the one or more virtual datacenters in a connectivity group. The routing table of this additional gateway router is configured with routes that specify (i) to send data traffic directed to the on-premises datacenter to that datacenter and (ii) to send data traffic directed to the virtual datacenters to the centralized gateway router configured for the connectivity group. Similarly, the routing table of the centralized gateway router is configured with routes that specify to send data traffic directed to the on-premises datacenter to the additional gateway.

(11) The centralized gateway router is also configured with routes to send traffic between any VPCs connected to the group and the virtual datacenters. Some embodiments place restrictions on the traffic that (i) allow traffic between two virtual datacenters in a group, (ii) allow traffic between any virtual datacenter in the group and a native VPC connected to the centralized gateway, (iii) allow traffic between any virtual datacenter in the group and an on-premises datacenter connected to the centralized gateway, (iv) do not allow traffic between two different on-premises datacenters connected to the centralized gateway, (v) do not allow traffic between two different native VPCs connected to the centralized gateway, and (vi) do not allow traffic between native VPCs and on-premises datacenters connected to the centralized gateways. In some embodiments, these are default security rules that the user of the cloud management platform can modify if desired.

(12) The preceding Summary is intended to serve as a brief introduction to some embodiments of the invention. It is not meant to be an introduction or overview of all inventive subject matter disclosed in this document. The Detailed Description that follows and the Drawings that are referred to in the Detailed Description will further describe the embodiments described in the Summary as well as other embodiments. Accordingly, to understand all the embodiments described by this document, a full review of the Summary, Detailed Description and the Drawings is needed. Moreover, the claimed subject matters are not to be limited by the illustrative details in the Summary, Detailed Description and the Drawings, but rather are to be defined by the appended claims, because the claimed subject matters can be embodied in other specific forms without departing from the spirit of the subject matters.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The novel features of the invention are set forth in the appended claims. However, for purpose of explanation, several embodiments of the invention are set forth in the following figures.
- (2) FIG. 1 conceptually illustrates a tenant logical network for a virtual datacenter of some embodiments.
- (3) FIG. 2 conceptually illustrates the physical implementation of the virtual datacenter in a public cloud datacenter according to some embodiments.
- (4) FIG. 3 conceptually illustrates a group of virtual datacenters connected in a connectivity group.
- (5) FIG. 4 conceptually illustrates a process of some embodiments for configuring a gateway router and connector routers in order to connect a set of virtual datacenters added to a connectivity group.
- (6) FIG. 5 conceptually illustrates the routing tables generated by a cloud management platform to configure the gateway router and connector routers for the connectivity group shown in FIG. 3.
- (7) FIG. 6 conceptually illustrates two virtual datacenters that are connected via a connectivity group to each other as well as to a native VPC and an on-premises datacenter.
- (8) FIG. 7 conceptually illustrates a group of virtual datacenters connected in a connectivity group that spans two geographic regions.
- (9) FIG. 8 conceptually illustrates an electronic system with which some embodiments of the invention are implemented.

DETAILED DESCRIPTION

- (10) In the following detailed description of the invention, numerous details, examples, and embodiments of the invention are set forth and described. However, it will be clear and apparent to one skilled in the art that the invention is not limited to the embodiments set forth and that the invention may be practiced without some of the specific details and examples discussed.
- (11) Some embodiments provide a method for providing connectivity between virtual datacenters defined and operating within public clouds. The virtual datacenters, in some embodiments, are datacenters that include network and/or compute management components operating in the public cloud as well as network endpoints (workloads) connected by a logical network within the virtual datacenter. The network endpoints as well as the logical network are managed by the management components that operate within the virtual datacenter.
- (12) To provide connectivity between these virtual datacenters, a cloud management platform enables a user (e.g., a network administrator of the enterprise to which the virtual datacenters belong) to define a connectivity group of virtual datacenters and add virtual datacenters to the group. In some embodiments, all of the virtual datacenters in a group are defined and operate within physical datacenters of the same public cloud provider, while in other embodiments the virtual datacenters may be defined in datacenters of multiple different public cloud providers.
- (13) In some embodiments, the virtual datacenters are defined within virtual private clouds (VPCs)

of the public cloud. A VPC, in some embodiments, is a set of workloads that are allocated to the tenant of the public cloud (e.g., an enterprise) and that are isolated from workloads of other tenants. In some embodiments, for a virtual datacenter, the tenant VPC is allocated a set of physical host computers of the public cloud that only host workload data compute nodes (e.g., virtual machines (VMs), containers, etc.) that are part of the tenant virtual datacenter (i.e., the physical host computers are not shared with other tenants of the public cloud). Within the VPC, a tenant logical network is defined, to which both the management components and the endpoint workloads connect.

(14) FIG. 1 conceptually illustrates a tenant logical network **100** for a virtual datacenter **105** of some embodiments. As shown, the logical network **100** of the virtual datacenter is defined to include a tier-0 (T0) logical router **110**, a management tier-1 (T1) logical router **115**, and a compute tier-1 (T1) logical router **120**. The T0 logical router **110** handles traffic entering and exiting the virtual datacenter **105** (e.g., traffic sent to other virtual datacenters or physical on-premises datacenters via a connectivity group, traffic between workloads at the virtual datacenter **105** and client devices connected to public networks, etc.). In addition, in some embodiments, some or all traffic between the management portion of the logical network **100** and the network endpoint workload data compute nodes (DCNs) is sent through the T0 logical router **110** (e.g., as a logical path between the management T1 logical router **115** and the compute T1 logical router **120**). The T0 logical router **110** may also be defined to provide certain services for data traffic that it processes.

(15) The management and compute T1 logical routers **115** and **120** are sometimes referred to as the management gateway and compute gateway. In some embodiments, a typical virtual datacenter is defined with these two T1 logical routers connected to a T0 logical router, which segregates the management network segments from the compute network segments to which workload DCNs connect. In general, public network traffic from external client devices would not be allowed to connect to the management network but would (in certain cases) be allowed to connect to the compute network (e.g., if the compute network includes web servers for a public-facing application). Each of the T1 logical routers **115** and **120** may also apply services to traffic that it processes, whether that traffic is received from the T0 logical router **110** or received from one of the network segments underneath the T1 logical router.

(16) In this example, the virtual datacenter logical network **100** includes three management logical switches **125-135** (also referred to as network segments) and two compute logical switches **140-145**. In this example, one or more compute manager DCNs **150** connect to the first management logical switch **125** and one or more network manager and controller DCNs **155** connect to the second management logical switch **130**. The DCNs shown here may be implemented in the public cloud as virtual machines (VMs), containers, or other types of machines, in different embodiments. In some embodiments, multiple compute manager DCNs **150** form a compute manager cluster connected to the logical switch **125**, while multiple network manager DCNs **155** form a management plane cluster and multiple network controller DCNs **155** form a control plane cluster (both of which are connected to the same logical switch **130**).

(17) The virtual datacenter **105** also includes workload DCNs **160**. These DCNs can host applications that are accessed by users (e.g., employees of the enterprise that owns and manages the virtual datacenter **105**), external client devices (e.g., individuals accessing a web server through a public network), or other DCNs (e.g., in the same virtual datacenter or different datacenters). The workload DCNs **160** in this example connect to two logical switches **140** and **145** (e.g., because they implement different tiers of an application, or different applications altogether). These DCNs **160** can communicate with each other, with workload DCNs in other datacenters, etc. via the interfaces connected to these compute logical switches **140** and **145**. In addition, in some embodiments (and as shown in this example), the workload DCNs **160** include a separate interface (e.g., in a different subnet) that connects to a management logical switch **135**. The workload DCNs

160 communicate with the compute and network management DCNs **150** and **155** via this logical switch **135**, without requiring this control traffic to be sent through the T0 logical router **110**.

(18) FIG. 2 conceptually illustrates the physical implementation of the virtual datacenter **105** in a public cloud datacenter **200** according to some embodiments. As mentioned above, a virtual datacenter such as that shown in FIG. 1 is implemented within a VPC of a public cloud datacenter in some embodiments. This figure shows that the virtual datacenter **105** is implemented in an isolated VPC **205** of the public cloud datacenter **200**. In some embodiments, this VPC **205** is allocated not just a set of VMs or other DCNs that execute on host computers managed by the public cloud provider and potentially shared with other tenants of the public cloud, but rather a set of host computers **210-220** of the public cloud datacenter **200**. This allows the management DCNs to manage the hypervisors and other software of the host computers **210-220** (e.g., so that these hypervisors implement the virtual datacenter logical network **100**).

(19) In different embodiments, the entire virtual datacenter may be implemented on a single host computer of the public datacenter **200** (which may host many VMs, containers, or other DCNs) or multiple different host computers. As shown, in this example, at least two host computers **210** and **215** execute workload and/or management VMs. Another host computer **220** executes a gateway datapath **225**. In some embodiments, this gateway datapath **225** implements a centralized component of the T0 logical router, such that all traffic between any external networks and the virtual datacenter is processed by the gateway datapath **225**. Additional details regarding logical routers and their implementation can be found in U.S. Pat. No. 9,787,605, which is incorporated herein by reference.

(20) Now that virtual datacenters have been described, connectivity groups of some embodiments for securely connecting these virtual datacenters without requiring traffic to traverse public networks will be described. FIG. 3 conceptually illustrates a group of virtual datacenters **305-315** connected in a connectivity group. When such a connectivity group is defined, some embodiments automatically define and configure a centralized gateway router for connecting the virtual datacenters, as well as routers at the edges of each of the virtual datacenters that connect to the gateway router.

(21) As shown, the connectivity group in FIG. 3 includes three virtual datacenters **305-315** that are implemented in three different private clouds **320-330** (e.g., VPCs within a public cloud). Each of these virtual datacenters **305-315** includes a T0 logical router for handling traffic entering and exiting the virtual datacenter, as well as compute and network managers that manage workload DCNs in the virtual datacenter (as shown in FIGS. 1 and 2). In this example, all three of the virtual datacenters **305-315** are located in different private cloud datacenters, though in some embodiments some or all of the virtual datacenters in a group can be located in the same physical datacenter.

(22) The private cloud datacenters **320-330** can be within different physical datacenters that all belong to the same public cloud provider (e.g., AWS, Azure, etc.), or datacenters belonging to two or more different public cloud providers (e.g., two AWS datacenters and one Azure datacenter). In addition, as described further below, some embodiments require that all of the virtual datacenters in a group be implemented in public cloud datacenters located in the same geographic region (e.g., the Western US), while other embodiments allow virtual datacenters in a group to be located in different geographic regions (e.g., across multiple continents).

(23) FIG. 3 also shows that, as a result of the creation of a connectivity group for connecting the virtual datacenters **305-315**, a gateway router **300** is configured to handle traffic between the virtual datacenters, and connector routers **335-345** are configured at each of these virtual datacenters. Each of the connector routers **335-345** connects to the T0 logical router (e.g., to an uplink interface of the T0 logical router) at its respective virtual datacenter **305-315**, as well to the gateway router **300**. Thus, any traffic from one virtual datacenter to another is routed from the host computer that implements the T0 logical router at the first virtual datacenter, to the connector router for that

virtual datacenter, to the gateway router **300**, to the connector router at the destination virtual datacenter, to the host computer that implements the T0 logical router at the destination virtual datacenter, and thereby to the rest of the logical network at that virtual datacenter. That is, the gateway router **300** acts as a hub for routing traffic between the virtual datacenters in a connectivity group. In some embodiments, the gateway router **300** is implemented as a horizontally scalable regional logical router. That is, the gateway router **300** can be implemented as a single device or scaled to multiple devices that act as a single router in some embodiments. These devices can be located at a single cloud or scaled to multiple clouds, in different embodiments. The connector routers **335-345** are implemented as logical routers at each virtual datacenter, in some embodiments. In some embodiments, the connector routers are part of the virtual datacenter, while in other embodiments the connector routers are implemented within the private cloud but outside of the virtual datacenter.

(24) In some embodiments, the gateway router and connector routers are configured by a cloud management platform that is used to (i) setup the virtual datacenters and (ii) form connectivity groups for the virtual datacenters. The cloud management platform, in some embodiments, receives configuration instructions from a user (e.g., a network administrator for an enterprise that manages the virtual datacenters) and translates these configuration instructions into configuration data for the virtual datacenters, the gateway router, the connector routers, etc. The configuration instructions include requests such as creation of virtual datacenters, creation of a connectivity group, addition of virtual datacenters to the connectivity group, removal of virtual datacenters from the connectivity group, etc.

(25) FIG. 4 conceptually illustrates a process **400** of some embodiments for configuring a gateway router and connector routers in order to connect a set of virtual datacenters added to a connectivity group. In some embodiments, the process **400** is performed by a cloud management platform. This cloud management platform can be an application that executes on a network administrator's computing device, or an application that the network administrator accesses (e.g., via a web browser, thin client application, etc.) in order to provide configuration instructions to the cloud management platform. The process **400** will be described in part by reference to FIG. 5, which illustrates routing tables provided to the gateway router **300** and connector routers **335-345** from FIG. 3.

(26) As shown, the process **400** begins by receiving (at **405**) a definition of a connectivity group and the addition of virtual datacenters to that connectivity group. In some embodiments, a connectivity group is allowed to include only a single virtual datacenter, but this serves little practical purpose. In some embodiments, a network administrator for an enterprise creates the connectivity group through a cloud management platform interface (e.g., a graphical user interface). In some embodiments, the cloud management platform allows the enterprise to create multiple such groups. In addition, as further described below, some embodiments require that the virtual datacenters in a connectivity group be located within the same geographic region (e.g., due to requirements of the public cloud(s) that host the virtual datacenters). Other embodiments do not place such restrictions on the virtual datacenters of a connectivity group.

(27) Based on the creation of the connectivity group, the process **400** defines (at **410**) a gateway router for connecting virtual datacenters in the group. In some embodiments, this gateway router is a logical router that is implemented by multiple physical entities (e.g., at each of the public cloud datacenters at which the virtual datacenters are located). In other embodiments, the gateway router is implemented at a single location (e.g., by a single virtual machine or other entity).

(28) The process **400** then dynamically generates (at **415**) routing tables for the connector routers at each virtual datacenter (i) to route traffic for other VDCs to the gateway router and (ii) to route for their virtual datacenter to the internal logical network of the virtual datacenter. In some embodiments, these latter routes specify to route the traffic to the T0 logical router at the virtual datacenter. The process **400** also generates (at **420**) a routing table for the gateway router to route

traffic for each virtual datacenter to the connector router at that virtual datacenter.

(29) FIG. 5 conceptually illustrates the routing tables generated by a cloud management platform 500 to configure the gateway router 300 and connector routers 335-345 for the connectivity group shown in FIG. 3. In some embodiments, the cloud management platform 500 generates these routing tables and configures the gateway router 300 and connector routers 335-345 by providing the routing tables to the entities implementing these routers.

(30) The routing table 505 for the gateway router 300 includes (i) routes for IP addresses associated with the first virtual datacenter 305 that specify to route traffic directed to these IP addresses to the connector router 335 for the first virtual datacenter 305, (ii) routes for IP addresses associated with the second virtual datacenter 310 that specify to route traffic directed to these IP addresses to the connector router 340 for the second virtual datacenter 310, and (iii) routes for IP addresses associated with the third virtual datacenter 315 that specify to route traffic directed to these IP addresses to the connector router 345 for the third virtual datacenter 315. Each of these may include various subnets, so that the routing table 505 includes multiple routes for each datacenter. For instance, in some embodiments, these IP addresses include both IP addresses associated with the workload DCNs as well as IP addresses associated with the management components at each virtual datacenter. These IP addresses, in some embodiments, are the addresses that the various T1 logical routers at each virtual datacenter are configured to advertise. For instance, a compute T1 logical router might advertise only a single virtual IP address that represents an entire subnet of private IP addresses, with the T1 logical router performing load balancing and network address translation (NAT).

(31) The routing table 510 for the connector router 335 at the first virtual datacenter 305 includes (i) routes for IP addresses associated with the first virtual datacenter 305 that specify to route traffic directed to these IP addresses to the T0 logical router at the first virtual datacenter 305 as well as (ii) routes for IP addresses associated with the second and third virtual datacenters 310 and 315 that specify to route traffic directed to these IP addresses to the gateway router 500. The routing table 515 for the connector router 340 at the second virtual datacenter 310 includes (i) routes for IP addresses associated with the second virtual datacenter 310 that specify to route traffic directed to these IP addresses to the T0 logical router at the second virtual datacenter 310 as well as (ii) routes for IP addresses associated with the first and third virtual datacenters 305 and 315 that specify to route traffic directed to these IP addresses to the gateway router 500. Lastly, the routing table 520 for the connector router 345 at the third virtual datacenter 315 includes (i) routes for IP addresses associated with the third virtual datacenter 315 that specify to route traffic directed to these IP addresses to the T0 logical router at the third virtual datacenter 315 as well as (ii) routes for IP addresses associated with the first and second virtual datacenters 305 and 310 that specify to route traffic directed to these IP addresses to the gateway router 500.

(32) It should also be noted that in some such embodiments, the T0 logical router at each virtual datacenter is configured (e.g., with static routes) to route traffic for the IP addresses associated with each of the other virtual datacenters in the connectivity group to the local connector router. In some embodiments, the cloud management platform provides this information to the network management and control system at each virtual datacenter in the group, allowing the network manager and controller components to configure the logical networking within the virtual datacenter.

(33) Returning to FIG. 4, the process 400 also dynamically generates (at 425) firewall rules to allow traffic between the virtual datacenters through the gateway router. In some embodiments, the default configuration for the gateway router and connector routers that enable a connectivity group is to block all traffic (as these routers should only be processing traffic for the specific network addresses associated with the virtual datacenters in the group). As such, some embodiments generate firewall rules for these routers allowing traffic with destination addresses associated with any of the virtual datacenters. In addition, some embodiments either automatically create firewall

rules for the logical networks at each of the virtual datacenters or require the administrator to separately configure these firewall rules in order to enable traffic to be sent between workload endpoint DCNs at the different datacenters within a group.

(34) Finally, the process **400** configures (at **430**) the gateway router and the connector routers. The process then ends. In some embodiments, these routers are provided by the public cloud provider(s) that host the virtual datacenters. In this case, some embodiments interface with the public cloud provider system in order for the public cloud provider to configure the gateway router and/or connector routers based on information provided by the cloud management platform.

(35) In addition, when networking at one of the virtual datacenters changes in a way that will affect the routes, the cloud management platform is responsible for modifying any of the affected routing tables in some embodiments. For instance, if a new subnet is added at a particular virtual datacenter by a network administrator (and advertised outside of that virtual datacenter), then the routing tables for all of the connector routers and the gateway router for the connectivity group to which the virtual datacenter belongs will be dynamically modified by the cloud management platform to account for that new subnet. Similarly, if a subnet is deleted at a virtual datacenter, then the cloud management platform dynamically modifies the routing tables for the gateway and connector routers to remove the routes for the deleted subnet. Along these lines, if a virtual datacenter is added to or removed from the group, then the cloud management platform dynamically adds or removes routes for all of the network addresses associated with that virtual datacenter (in addition to adding or removing the connector router for the virtual datacenter).

(36) In the example shown in FIGS. **3** and **5**, the connectivity group only connects virtual datacenters. In some embodiments, the cloud management platform also allows the user to connect the virtual datacenters in a group to (i) native virtual private clouds in a public cloud and/or (ii) on-premises datacenters. The native VPCs, in some embodiments, are not virtual datacenters in that they do not include management components and, in many cases, are not implemented on fully isolated hardware. That is, the host computers that host DCNs of the VPC may also host DCNs for other VPCs of other public cloud tenants. In some embodiments, the native VPCs may host services (e.g., load balancers, deep packet inspection, DHCP service, authentication service, endpoint service, etc.) used to process data traffic sent to or from the virtual datacenters. In some embodiments, these services in the native VPCs can include any custom service defined by the enterprise and running on a native cloud instance.

(37) In addition, these native VPCs may be managed through the cloud management platform or through a cloud provider platform (i.e., the platform for the cloud provider to which the VPC is native) in different embodiments. For example, some embodiments enable the administrator for the enterprise to link an account with the cloud management platform that creates and manages the connectivity groups with the platform of a specific cloud provider that hosts the native VPCs (and, in some cases, the gateway router for the connectivity group). This allows the administrator to manage the native VPCs and attach these native VPCs to the gateway router for the connectivity group.

(38) An administrator might also want the virtual datacenters to be able to connect to physical on-premises datacenters (e.g., a primary on-premises datacenter and one or more branch office datacenters). For instance, if only a portion of the enterprise's workload DCNs are moved into virtual datacenters, the on-premises workload DCNs might need to communicate with the workload DCNs in the virtual datacenter.

(39) FIG. **6** conceptually illustrates two virtual datacenters **605** and **610** that are connected via a connectivity group to each other as well as to a native VPC **615** and an on-premises datacenter **620**. As shown, connector routers **625** and **630** are configured at each of the two virtual datacenters **605** and **610**, and these connect to a gateway router **600** as in the example of FIG. **3**. The native VPC **615**, in some embodiments, connects to the gateway router **600** via a VPC attachment when the native VPC and the gateway router are provided by the same public cloud provider. In other

embodiments, an additional intervening router (similar to the connector routers **625** and **630**) is required in order to connect the native VPC **615** to the virtual datacenters **605** and **610**.

(40) To connect the on-premises datacenter **620**, in some embodiments an additional connector gateway router **635** is defined (and, e.g., implemented in the public cloud). Whereas the virtual datacenters **605** and **610**, the native VPC **615**, and the gateway router **600** all operate in public cloud datacenters (possibly belonging to the same public cloud provider), the on-premises datacenter **620** is a physical datacenter belonging to the enterprise. The connector gateway **635** is a specialized on-premises connection gateway provided by the public cloud provider in some embodiments, which attaches to the gateway router **600** and transmits traffic between the gateway router **600** and the on-premises datacenter **620**. In different embodiments, the administrator defines the connector gateway **635** either directly through the cloud provider platform or through the cloud management platform (which interfaces with the cloud provider platform). The administrator can then connect the connector gateway **635** to the gateway router **600** in order to connect the on-premises datacenter **620** to the virtual datacenters **605** and **610** of the connectivity group.

(41) FIG. **6** also shows the routing tables for the gateway router **600**, the connector routers **625** and **630**, the native VPC **615** (i.e., a router at the native VPC responsible for connecting to the gateway router **600**), and the connector gateway **635**. The routing table **640** for the connector router **625** at the first virtual datacenter **605** includes (i) routes for IP addresses associated with the first virtual datacenter **605** that specify to route traffic directed to these IP addresses to the T0 logical router at the first virtual datacenter **605** as well as (ii) routes for IP addresses associated with the second virtual datacenter **610**, the native VPC **615**, and the on-premises datacenter **620** that specify to route traffic directed to these IP addresses to the gateway router **600**. Similarly, the routing table **645** for the connector router **630** at the second virtual datacenter **610** includes (i) routes for IP addresses associated with the second virtual datacenter **610** that specify to route traffic directed to these IP addresses to the T0 logical router at the second virtual datacenter **610** as well as (ii) routes for IP addresses associated with the first virtual datacenter **605**, the native VPC **615**, and the on-premises datacenter **620** that specify to route traffic directed to these IP addresses to the gateway router **600**.

(42) In turn, the routing table **650** for the gateway router **600** includes (i) routes for IP addresses associated with the first virtual datacenter **605** that specify to route traffic directed to these IP addresses to the connector router **625** for the first virtual datacenter **605**, (ii) routes for IP addresses associated with the second virtual datacenter **610** that specify to route traffic directed to these IP addresses to the connector router **630** for the second virtual datacenter **610**, (iii) routes for IP addresses associated with the native VPC **615** that specify to route traffic directed to these IP addresses to the attachment for the native VPC **615**, and (iv) routes for IP addresses associated with the on-premises datacenter **620** that specify to route traffic directed to these IP addresses to the connector gateway **635**.

(43) The routing table **655** for the native VPC attachment includes (i) routes for IP addresses associated with the native VPC **615** that specify to route traffic directed to these IP addresses internally and (ii) routes for IP addresses associated with the first and second virtual datacenters **605** and **610** that specify to route traffic directed to these IP addresses to the gateway router **600**. Similarly, the routing table **660** for the connector gateway **635** includes (i) routes for IP addresses associated with the on-premises datacenter **620** that specify to route traffic directed to these IP addresses to a router at that datacenter and (ii) routes for IP addresses associated with the first and second virtual datacenters **605** and **610** that specify to route traffic directed to these IP addresses to the gateway router **600**.

(44) In addition, the routing table **655** for the native VPC attachment indicates that traffic directed to IP addresses associated with the on-premises datacenter should be dropped. Similarly, the routing table **660** for the connector gateway **635** indicates that traffic directed to IP addresses associated with the native VPC should be dropped. It should be noted that in some embodiments

these rules are implemented as firewall rules rather than as routes in the routing tables. Some embodiments place restrictions on the traffic that (i) allow traffic between two virtual datacenters in a group, (ii) allow traffic between any virtual datacenter in the group and a native VPC connected to the centralized gateway router, (iii) allow traffic between any virtual datacenter in the group and an on-premises datacenter connected to the centralized gateway, (iv) do not allow traffic between two different on-premises datacenters connected to the centralized gateway, (v) do not allow traffic between two different native VPCs connected to the centralized gateway, and (vi) do not allow traffic between native VPCs and on-premises datacenters connected to the centralized gateways. In some embodiments, these are default security rules that the user of the cloud management platform can modify if desired.

(45) In addition, as mentioned above, some embodiments impose a restriction that all of the virtual datacenters in a connectivity group must be located in datacenters within a single predefined geographic region (e.g., Eastern US, Western US, Europe, East Asia, etc.). These geographic regions can be defined by the cloud management platform through which the groups are defined, or by the public cloud provider(s) that host the virtual datacenters.

(46) Other embodiments allow for a connectivity group to include virtual datacenters from multiple such geographic regions. To include virtual datacenters from multiple regions, some embodiments require gateway routers to be defined and configured in each of the geographic regions, with these gateway routers connected to each other in order to enable connectivity between all of the virtual datacenters.

(47) FIG. 7 conceptually illustrates a group of virtual datacenters **705-720** connected in a connectivity group that spans two geographic regions **725** and **730**. In this case, when the virtual datacenters in a connectivity group span two or more regions, some embodiments define a gateway router in each geographic region rather than a single gateway router to which all of the virtual datacenters connect. In this case, the two virtual datacenters **705** and **710** located in the first geographic region **725** connect to a first gateway router **735**, while the two virtual datacenters **715** and **720** located in the second geographic region **730** connect to a second gateway router **740**. In some embodiments, if any native VPCs are connected to the group, these connect to the gateway router in their geographic region. Similarly, to connect an on-premises datacenter to the group, a connector gateway is instantiated in a particular region that connects to the gateway router in that particular region.

(48) The routing tables **765-780** for connector routers **745-760** at each of the virtual datacenters specify to send incoming traffic for the local virtual datacenter to the internal logical network of that virtual datacenter and to send outgoing traffic for any of the other virtual datacenters to the local gateway router. Specifically, the routing table **765** for the connector router **745** at the first virtual datacenter **705** includes (i) routes for IP addresses associated with the first virtual datacenter **705** that specify to route traffic directed to these IP addresses to the T0 logical router at the first virtual datacenter **705** as well as (ii) routes for IP addresses associated with the second, third, and fourth virtual datacenters **710-720** that specify to route traffic directed to these IP addresses to the first gateway router **735**. The routing table **770** for the connector router **750** at the second virtual datacenter **710** includes (i) routes for IP addresses associated with the second virtual datacenter **710** that specify to route traffic directed to these IP addresses to the T0 logical router at the second virtual datacenter **710** as well as (ii) routes for IP addresses associated with the first, third, and fourth virtual datacenters **705**, **715**, and **720** that specify to route traffic directed to these IP addresses to the first gateway router **735**. The routing table **775** for the connector router **755** at the third virtual datacenter **715** includes (i) routes for IP addresses associated with the third virtual datacenter **715** that specify to route traffic directed to these IP addresses to the T0 logical router at the third virtual datacenter **715** as well as (ii) routes for IP addresses associated with the first, second, and fourth virtual datacenters **705**, **710**, and **720** that specify to route traffic directed to these IP addresses to the second gateway router **740**. Finally, the routing table **780** for the connector

router **760** at the fourth virtual datacenter **720** includes (i) routes for IP addresses associated with the fourth virtual datacenter **720** that specify to route traffic directed to these IP addresses to the T0 logical router at the fourth virtual datacenter **720** as well as (ii) routes for IP addresses associated with the first, second, and third virtual datacenters **705-715** that specify to route traffic directed to these IP addresses to the second gateway router **740**.

(49) In each region, the gateway router is configured to send traffic for the virtual datacenters in that region to the connector router at that virtual datacenter and to send traffic for the virtual datacenters in other regions to the gateway router in the region at which the destination virtual datacenter is located. Specifically, the routing table **785** for the first gateway router **735** in the first region **725** includes (i) routes for IP addresses associated with the first virtual datacenter **705** that specify to route traffic directed to these IP addresses to the connector router **745** for the first virtual datacenter **705**, (ii) routes for IP addresses associated with the second virtual datacenter **710** that specify to route traffic directed to these IP addresses to the connector router **750** for the second virtual datacenter **710**, and (iii) routes for IP addresses associated with the third and fourth virtual datacenters **715** and **720** that specify to route traffic directed to these IP addresses to the second gateway router **740** in the second region **730**. Correspondingly, the routing table **790** for the second gateway router **740** in the second region **730** includes (i) routes for IP addresses associated with the third virtual datacenter **715** that specify to route traffic directed to these IP addresses to the connector router **755** for the third virtual datacenter **715**, (ii) routes for IP addresses associated with the fourth virtual datacenter **720** that specify to route traffic directed to these IP addresses to the connector router **760** for the second virtual datacenter **720**, and (iii) routes for IP addresses associated with the first and second virtual datacenters **705** and **710** that specify to route traffic directed to these IP addresses to the first gateway router **735** in the first region **725**.

(50) In some embodiments, when a group includes virtual datacenters located in more than two regions, the gateway routers form a mesh network. In other embodiments, not all of the gateway routers will connect directly to all of the other gateway routers. For instance, a connectivity group with virtual datacenters located in Western Europe, Eastern Europe, North Africa, Eastern US, and Western US might have a connection between the gateway routers in the two US regions, a mesh connection between the gateway routers in Europe and North Africa, and a connection between the Eastern US and Western Europe.

(51) FIG. **8** conceptually illustrates an electronic system **800** with which some embodiments of the invention are implemented. The electronic system **800** may be a computer (e.g., a desktop computer, personal computer, tablet computer, server computer, mainframe, a blade computer etc.), phone, PDA, or any other sort of electronic device. Such an electronic system includes various types of computer readable media and interfaces for various other types of computer readable media. Electronic system **800** includes a bus **805**, processing unit(s) **810**, a system memory **825**, a read-only memory **830**, a permanent storage device **835**, input devices **840**, and output devices **845**.

(52) The bus **805** collectively represents all system, peripheral, and chipset buses that communicatively connect the numerous internal devices of the electronic system **800**. For instance, the bus **805** communicatively connects the processing unit(s) **810** with the read-only memory **830**, the system memory **825**, and the permanent storage device **835**.

(53) From these various memory units, the processing unit(s) **810** retrieve instructions to execute and data to process in order to execute the processes of the invention. The processing unit(s) may be a single processor or a multi-core processor in different embodiments.

(54) The read-only-memory (ROM) **830** stores static data and instructions that are needed by the processing unit(s) **810** and other modules of the electronic system. The permanent storage device **835**, on the other hand, is a read-and-write memory device. This device is a non-volatile memory unit that stores instructions and data even when the electronic system **800** is off. Some embodiments of the invention use a mass-storage device (such as a magnetic or optical disk and its corresponding disk drive) as the permanent storage device **835**.

(55) Other embodiments use a removable storage device (such as a floppy disk, flash drive, etc.) as the permanent storage device. Like the permanent storage device **835**, the system memory **825** is a read-and-write memory device. However, unlike storage device **835**, the system memory is a volatile read-and-write memory, such a random-access memory. The system memory stores some of the instructions and data that the processor needs at runtime. In some embodiments, the invention's processes are stored in the system memory **825**, the permanent storage device **835**, and/or the read-only memory **830**. From these various memory units, the processing unit(s) **810** retrieve instructions to execute and data to process in order to execute the processes of some embodiments.

(56) The bus **805** also connects to the input and output devices **840** and **845**. The input devices enable the user to communicate information and select commands to the electronic system. The input devices **840** include alphanumeric keyboards and pointing devices (also called “cursor control devices”). The output devices **845** display images generated by the electronic system. The output devices include printers and display devices, such as cathode ray tubes (CRT) or liquid crystal displays (LCD). Some embodiments include devices such as a touchscreen that function as both input and output devices.

(57) Finally, as shown in FIG. **8**, bus **805** also couples electronic system **800** to a network **865** through a network adapter (not shown). In this manner, the computer can be a part of a network of computers (such as a local area network (“LAN”), a wide area network (“WAN”), or an Intranet, or a network of networks, such as the Internet. Any or all components of electronic system **800** may be used in conjunction with the invention.

(58) Some embodiments include electronic components, such as microprocessors, storage and memory that store computer program instructions in a machine-readable or computer-readable medium (alternatively referred to as computer-readable storage media, machine-readable media, or machine-readable storage media). Some examples of such computer-readable media include RAM, ROM, read-only compact discs (CD-ROM), recordable compact discs (CD-R), rewritable compact discs (CD-RW), read-only digital versatile discs (e.g., DVD-ROM, dual-layer DVD-ROM), a variety of recordable/rewritable DVDs (e.g., DVD-RAM, DVD-RW, DVD+RW, etc.), flash memory (e.g., SD cards, mini-SD cards, micro-SD cards, etc.), magnetic and/or solid state hard drives, read-only and recordable Blu-Ray® discs, ultra-density optical discs, any other optical or magnetic media, and floppy disks. The computer-readable media may store a computer program that is executable by at least one processing unit and includes sets of instructions for performing various operations. Examples of computer programs or computer code include machine code, such as is produced by a compiler, and files including higher-level code that are executed by a computer, an electronic component, or a microprocessor using an interpreter.

(59) While the above discussion primarily refers to microprocessor or multi-core processors that execute software, some embodiments are performed by one or more integrated circuits, such as application specific integrated circuits (ASICs) or field programmable gate arrays (FPGAs). In some embodiments, such integrated circuits execute instructions that are stored on the circuit itself.

(60) As used in this specification, the terms “computer”, “server”, “processor”, and “memory” all refer to electronic or other technological devices. These terms exclude people or groups of people. For the purposes of the specification, the terms display or displaying means displaying on an electronic device. As used in this specification, the terms “computer readable medium,” “computer readable media,” and “machine readable medium” are entirely restricted to tangible, physical objects that store information in a form that is readable by a computer. These terms exclude any wireless signals, wired download signals, and any other ephemeral signals.

(61) This specification refers throughout to computational and network environments that include virtual machines (VMs). However, virtual machines are merely one example of data compute nodes (DCNs) or data compute end nodes, also referred to as addressable nodes. DCNs may include non-virtualized physical hosts, virtual machines, containers that run on top of a host operating system

without the need for a hypervisor or separate operating system, and hypervisor kernel network interface modules.

(62) VMs, in some embodiments, operate with their own guest operating systems on a host using resources of the host virtualized by virtualization software (e.g., a hypervisor, virtual machine monitor, etc.). The tenant (i.e., the owner of the VM) can choose which applications to operate on top of the guest operating system. Some containers, on the other hand, are constructs that run on top of a host operating system without the need for a hypervisor or separate guest operating system. In some embodiments, the host operating system uses name spaces to isolate the containers from each other and therefore provides operating-system level segregation of the different groups of applications that operate within different containers. This segregation is akin to the VM segregation that is offered in hypervisor-virtualized environments that virtualize system hardware, and thus can be viewed as a form of virtualization that isolates different groups of applications that operate in different containers. Such containers are more lightweight than VMs.

(63) Hypervisor kernel network interface modules, in some embodiments, is a non-VM DCN that includes a network stack with a hypervisor kernel network interface and receive/transmit threads. One example of a hypervisor kernel network interface module is the vmknix module that is part of the ESXi™ hypervisor of VMware, Inc.

(64) It should be understood that while the specification refers to VMs, the examples given could be any type of DCNs, including physical hosts, VMs, non-VM containers, and hypervisor kernel network interface modules. In fact, the example networks could include combinations of different types of DCNs in some embodiments.

(65) While the invention has been described with reference to numerous specific details, one of ordinary skill in the art will recognize that the invention can be embodied in other specific forms without departing from the spirit of the invention. In addition, a number of the figures (including FIG. 4) conceptually illustrate processes. The specific operations of these processes may not be performed in the exact order shown and described. The specific operations may not be performed in one continuous series of operations, and different specific operations may be performed in different embodiments. Furthermore, the process could be implemented using several sub-processes, or as part of a larger macro process. Thus, one of ordinary skill in the art would understand that the invention is not to be limited by the foregoing illustrative details, but rather is to be defined by the appended claims.

Claims

1. A method for routing data traffic between a group of virtual datacenters implemented in a public cloud, the method comprising: at a gateway router to which each of at least two of the virtual datacenters in the group of virtual datacenters connect, each respective virtual datacenter comprising a respective set of network management components implemented in the public cloud and a respective set of network endpoints executing in the public cloud and connected by a respective logical network that is managed by the respective network management components of respective virtual datacenter: receiving routes advertised by the respective logical network at each of the virtual datacenters that connect to the gateway router; and using the received routes to route data traffic, sent between the logical networks of the virtual datacenters of the group, to routers configured between the logical networks at each of the virtual datacenters and the gateway router; wherein first traffic entering and exiting a virtual datacenter is associated with a first tier router and a second traffic between the network management components is associated with a second tier router, the first tier router and the second tier router being configured to separate the first traffic and the second traffic, the first tier router being a tier-0 router, the second tier router being a tier-1 router, a virtual datacenter being defined with two tier-1 routers connected to a tier-0 router, the tier-0 router comprising a logical router configured to handle traffic entering and exiting the virtual

datacenter, the two tier-1 routers comprising a management gateway and a compute gateway segregated by the tier-0 router.

2. The method of claim 1, wherein the routes for each virtual datacenter comprise (i) routes for network addresses associated with the network management components of the virtual datacenter and (ii) routes for network addresses associated with the logical network endpoints of the virtual datacenter.

3. The method of claim 1, wherein (i) the gateway router and (ii) the routers configured for each of the virtual datacenters to connect the virtual datacenters to the gateway router are configured by a cloud management platform.

4. The method of claim 1, wherein the logical network at a particular virtual datacenter comprises: a first logical router for handling data traffic entering and exiting the particular virtual datacenter; a second logical router for connecting one or more management network segments to the first logical router, wherein the network management components are connected to the management network segments; and a third logical router for connecting one or more logical network segments with network endpoints to the first logical router.

5. The method of claim 4, wherein the routes received from the particular datacenter are advertised by the second and third logical routers.

6. The method of claim 1 further comprising: receiving routes for network addresses associated with a virtual private cloud (VPC) of the public cloud that hosts services native to the public cloud; and using the routes for the network addresses associated with the VPC to route data traffic, sent from the logical networks of the virtual datacenters to the services hosted at the VPC, to a network attachment for the VPC.

7. The method of claim 1, wherein the group is associated with a particular enterprise, wherein the gateway router is a first gateway router, wherein the method further comprises: receiving routes for network addresses associated with an on-premises datacenter of the enterprise that belongs to the group; and using the routes for the network addresses associated with the on-premises datacenter to route data traffic, sent from the logical networks of the virtual datacenters to network endpoints located at the on-premises datacenter, to a second gateway router that provides connectivity for the on-premises datacenter.

8. The method of claim 7, wherein the second gateway is implemented in the public cloud and uses one of a virtual private network (VPN) and a dedicated line to communicate with the on-premises datacenter.

9. The method of claim 7, wherein the first gateway router (i) allows data traffic between the virtual datacenters and the on-premises datacenter, (ii) allows data traffic between the virtual datacenters and any VPCs added to the group, (iii) does not allow traffic between the on-premises datacenter and any other on-premises datacenters added to the group, and (iv) does not allow traffic between the on-premises datacenter and any VPCs added to the group.

10. The method of claim 1, wherein the gateway router and the at least two datacenters are located in a first region of the public cloud, wherein the gateway router is a first gateway router, wherein the method further comprises: receiving routes advertised by the respective logical networks at each of a set of one or more virtual datacenters that are located in a second region of the public cloud and connect to a second gateway router also located in the second region of the public cloud; and using said routes to route data traffic, sent from the logical networks of the virtual datacenters located in the first region to the logical networks of the virtual datacenters located in the second region, to the second gateway router.

11. The method of claim 1, wherein the gateway router is implemented as a horizontally scalable router in the public cloud.

12. A non-transitory machine-readable medium storing a program which when executed by at least one processing unit implements a gateway router for routing data traffic between a group of virtual datacenters implemented in a public cloud, the program comprising sets of instructions for:

receiving routes advertised by a respective logical network at each of at least two virtual datacenters in the group of virtual datacenters that connect to the gateway router, each respective virtual datacenter comprising a respective set of network management components implemented in the public cloud and a respective set of network endpoints executing in the public cloud and connected by a respective logical network that is managed by the respective network management components of respective virtual datacenter; and using the received routes to route data traffic, sent between the logical networks of the virtual datacenters of the group, to routers configured between the logical networks at each of the virtual datacenters and the gateway router; wherein a first traffic entering and existing a virtual datacenter is associated with a first tier router and a traffic between the network management components is associated with a second tier router, the first tier router and the second tier router being configured to separate the first traffic and the second traffic, the first tier router being a tier-0 router, the second tier router being a tier-1 router, a virtual datacenter being defined with two tier-1 routers connected to a tier-0 router, the tier-0 router comprising a logical router configured to handle traffic entering and exiting the virtual datacenter, the two tier-1 routers comprising a management gateway and a compute gateway segregated by the tier-0 router.

13. The non-transitory machine-readable medium of claim 12, wherein the routes for each virtual datacenter comprise (i) routes for network addresses associated with the network management components of the virtual datacenter and (ii) routes for network addresses associated with the logical network endpoints of the virtual datacenter.

14. The non-transitory machine-readable medium of claim 12, wherein (i) the gateway router and (ii) the routers configured for each of the virtual datacenters to connect the virtual datacenters to the gateway router are configured by a cloud management platform.

15. The non-transitory machine-readable medium of claim 12, wherein the logical network at a particular virtual datacenter comprises: a first logical router for handling data traffic entering and exiting the particular virtual datacenter; a second logical router for connecting one or more management network segments to the first logical router, wherein the network management components are connected to the management network segments; and a third logical router for connecting one or more logical network segments with network endpoints to the first logical router.

16. The non-transitory machine-readable medium of claim 12, wherein the program further comprises sets of instructions for: receiving routes for network addresses associated with a virtual private cloud (VPC) of the public cloud that hosts services native to the public cloud; and using the routes for the network addresses associated with the VPC to route data traffic, sent from the logical networks of the virtual datacenters to the services hosted at the VPC, to a network attachment for the VPC.

17. The non-transitory machine-readable medium of claim 12, wherein the group is associated with a particular enterprise, wherein the gateway router is a first gateway router, wherein the program further comprises sets of instructions for: receiving routes for network addresses associated with an on-premises datacenter of the enterprise that belongs to the group; and using the routes for the network addresses associated with the on-premises datacenter to route data traffic, sent from the logical networks of the virtual datacenters to network endpoints located at the on-premises datacenter, to a second gateway router that provides connectivity for the on-premises datacenter.

18. The non-transitory machine-readable medium of claim 17, wherein the second gateway is implemented in the public cloud and uses one of a virtual private network (VPN) and a dedicated line to communicate with the on-premises datacenter.

19. The non-transitory machine-readable medium of claim 17, wherein the first gateway router (i) allows data traffic between the virtual datacenters and the on-premises datacenter, (ii) allows data traffic between the virtual datacenters and any VPCs added to the group, (iii) does not allow traffic between the on-premises datacenter and any other on-premises datacenters added to the group, and (iv) does not allow traffic between the on-premises datacenter and any VPCs added to the group.

20. The non-transitory machine-readable medium of claim 2, wherein the gateway router and the at

least two datacenters are located in a first region of the public cloud, wherein the gateway router is a first gateway router, wherein the program further comprises sets of instructions for: receiving routes advertised by the respective logical networks at each of a set of one or more virtual datacenters that are located in a second region of the public cloud and connect to a second gateway router also located in the second region of the public cloud; and using said routes to route data traffic, sent from the logical networks of the virtual datacenters located in the first region to the logical networks of the virtual datacenters located in the second region, to the second gateway router.
